CHAPTER 15

CITIES THAT THINK

ARCHITECTURE IN AN EMERGING CULTURE

We recommend that the present state and aim of architecture also be redefined to fit the needs of an emerging future. The questions we should concern ourselves with are: *What ends are these new cities to serve?* and *What are the prime considerations in designing a place of residence?*

In simplified form, a home may be any type of enclosure that protects people from varying weather conditions and provides for most of the occupant's primary needs: a place to rest, sleep, work, and carry out the business of ordinary living. We presently think of a shelter or dwelling as a suitable structure fabricated of wood, steel, concrete, and glass or a combination of materials. We envision a structure with windows to provide light and exterior walls for privacy. We install bathrooms for sanitary purposes. We utilize electricity for heating, air-conditioning and so forth. Our notions about a home generally reflect these limited concepts.

Throughout history, dwellings took many forms. People sought shelter from the effects of weather in caves. Others used a wigwam, a lean-to, and floating habitats. All manner of materials were utilized, including bamboo, clay, the sides of cliffs and hills, domes of ice, and countless others. At present people are seriously considering the possible colonization of the sea and outer space. Since shelters take so many forms, we have to broaden our expectations of shelters. Although one does not ordinarily think of an early diving suit as a shelter, it protects the wearer from the immediate surroundings, that is, from the hostile element of the sea. A space suit provides similar protection. Such suits enable people to function in an environment that does not ordinarily support life. From body enclosures to single dwellings, multiple dwellings, and eventually to total enclosure systems in which the entire city works together as a single organism — this could be the evolution of shelters.

In times to come, the effects of weather may be shielded by electronic means. The furnishings of future dwellings may consist of totally different configurations that automatically adjust to our body contours. Newer technologies may render walls entirely transparent so that occupants can view the surrounding landscape without anyone on the outside being able to see in. Daylight could be softened and subdued according to the preference of the occupants. These buildings would provide a barrier to sound, insects, and dust, and maintain the desired, optimal internal temperature. The telephone of the future might

not have the physical appearance we are familiar with: it may be entirely invisible and a component part of the interior structure. It may focus sound to the proximity of your ear by electronic means. The building's materials may be energy generating and control their own, surrounding climate.

If we look upon such a dwelling with our present habits of thought, it appears unfamiliar and very different from what we are accustomed to. It may even appear, to some, to be of a surreal nature.

It is not so much that this new residence does not resemble a home, as one knows it; it is rather that it is foreign to our concept of what a home ought to be, as we understand it. We conceive our homes only within the restrictions of our own habits of thought and indoctrination.

In times to come, the definition of anything in the physical world will not be as restricted to appearance only; rather, it will be thought of in terms of the function it renders.

While some advocate the modification of existing cities and spend much time and resources attempting to update them, we find such attempts inadequate. Modification carries a large price in dollars and sustainment resources. Modifying and building on what we have means we must continue to support a combination of older systems' infrastructure and energy needs, their high cost of operation and maintenance, and their overall inefficiency, not to mention their detrimental effect on the occupants. It is actually less expensive to build newer cities from the ground up than to restore and maintain the old ones, just as it is far more efficient and less costly to design a flexible, state of the art production method than to attempt to upgrade an obsolete factory.

To live in a world without pollution and waste, yet keep parks, playgrounds, art and music centers, schools, and health care available to everyone without a price tag, profound changes are required in the way we plan our cities and conduct human affairs.

The innovative, multi-dimensional and circular cities we propose combine the most sophisticated utilization of available resources and construction techniques. The geometrically elegant, circular arrangement, surrounded by parks and gardens, is designed to operate with the minimum expenditure of energy in order to obtain the highest possible standard of living for everyone. The city would use the best of clean technology in harmony with the local ecology.

The design and development of these new cities emphasize the restoration and protection of the environment: in our view, technology without human concern is meaningless.

In a resource-based economy the circular arrangement employs a systems approach, efficiently applying resources and energy conservation, ease of fabrication and relative freedom from maintenance. The process of assembling entire cities through the standardization of basic, structural systems prefabricated in automated plants and often assembled on site, is the only feasible way to provide a very high standard of living for all people within the shortest possible time. This permits a wide range of flexibility in design and takes advan-

tage of interchangeable units. In this way cities could take on new and different appearances depending on their function. Each would be unique. This approach does not reduce the lives of people to a subsistence level; rather, it makes available all the amenities that modern science and technology can provide. Even the wealthiest people of today could not achieve a standard of living equal to that of a resource-based economy.

Prefab modular units could converge on a building site to facilitate automatic assembly. All the electronics could be an integral part of the building of modular components, each one easily connected to existing power supplies — if the buildings are not entirely self-generating. All construction would be prescheduled to minimize interference with ongoing traffic.

The shape of tomorrow's cities — their geometry, external appearance, and the total configuration — would be a direct expression of the function they are designed to serve, keeping in mind that such a city is the extension of human activity in complete harmony with the surrounding environment. These new cities would serve as universities. Each would be designed to carry out specific, integrated functions and have a flexibility to permit changes for new and innovative installations. The size of the cities would vary depending on their geographic location and their use.

In the planning of new cities, computers would help determine the design parameters based upon the most appropriate environment to meet human and environmental needs. These total environments could be constructed to permit the widest possible range of individuality and creativity for the inhabitants.

We are often asked: Who will direct and program this cybernated city system? No-one will. The major difference between today's computer technology and the system we recommend is that our system extends its autonomic nervous system (environmental sensors) into all areas of the social complex. They function to coordinate a balance between production and distribution and operate a balanced-load economy. The decisions are arrived at on the basis of feedback from the environment.

For example, in the agricultural belt electronic probes embedded in the soil automatically maintain a constant inventory of the water table, soil conditions, nutrients, etc., and act appropriately without the need for human intervention as conditions change. This method of industrial electronic feedback could be applied to the entire system.

Another difference we propose is the reworking of concepts that drive our production planning. Instead of large, in-house facilities that create standardized products with limited application, we propose production devices that operate during the act of delivery. For example, transportation modules for ships, trains and planes that process time-sensitive products like fish and vegetables en route. The development of building materials that retain an element of fluidity would lead to the creation of indestructible houses, music centers, art centers and multi-purpose buildings in a variety of shapes and sizes.

What the new city would provide is a total environment, with clean air and water, health care, good nutrition, access to information and education for all. The city would have

art and music centers, fully equipped machine shops, science labs, hobby and sports areas, and manufacturing districts. These new cities could also provide all manners of recreation within a short distance of the residential district. This type of technology is inevitable. Waste recycling, renewable and clean power generating systems, and all manner of services would be managed by integrated, cybernated methods. The management of human affairs, such as life styles and personal preferences, are totally selected by the individual.

Some of the cities could be circular, while others may be linear, underground, or constructed as floating cities in the sea. Cables and satellite positioning could be incorporated to prevent drifting for sea-borne cities.

Eventually, many cities may be designed as total enclosure systems, much like a cruise ship outfitted for a six-month cruise. They would contain residences, theaters, parks, recreation, entertainment centers, health care and educational facilities, and all the requirements for a total living environment. Everything built in these cities would be as near to a self-contained system as conditions allow. In northern locations, some could be partially underground.

The underground or subterranean city of the future could be a total-enclosure dome system. The purpose of the first, subterranean mega city might be to research the possibilities of sustaining life forms on inhospitable planets. Many underground cities could be built in inhospitable regions of our planet. They can provide an ideal climate year round with lush gardens and waterfalls. In other words, they would be able to supply all the benefits of a subtropical environment. Some of these cities would be self-sustaining and possibly generate energy from geothermal sources.

The future will generate newer materials and methods, which will result in vastly different expressions of structure, form, and function, each consistent with an evolving and changing world. Most of the new materials will probably serve multiple purposes. They could be lightweight, high strength, and low maintenance, with acoustical properties not found in today's structures. These newer materials could combine all of these factors as an integral part of the structural components.

Some new structural materials may be of a sandwich-type construction that is semi-flexible, with an inner foam core and a glazed ceramic, outer surface to allow for expansion and contraction without fracture. This will require no maintenance. Their thin shell construction can be mass-produced in a matter of hours. With this type of construction, there would be minimal damage from earthquakes, hurricanes, termites, and fires. Windows would be controlled electronically to shade or darken external illumination and come equipped with computer-controlled, automatic cleaning systems that require no human labor.

Combining innovative technologies makes it possible to conserve resources for lesser-developed regions, without sacrificing any of the conveniences of advanced living. It is only through such a process of applying innovation that our end goals of a high standard of living for the entire human race can be achieved.

We have no shortage of material. The misuse and waste of resources by our monetary-oriented society create artificial scarcity.

When we look at a city as a biome that grows and adapts, requiring energy, food and water, disposal systems, and arteries for transportation of goods and people, our ideas of space and permanency change dramatically. Our present haphazard growth patterns reflect available space and access rather than any cohesive planning. Joining the city elements together in a predetermined way conserves energy and provides easy access to all portions of the city. Most of the prefabricated elements that comprise the city would be designed to permit modification as needed. With the introduction of newer materials, the city design can be continuously updated, while taking into consideration new technological and structural progress and evolving human patterns. All systems would be of an emergent nature and constructed to allow the maximum latitude in accommodating change. This could permit the city to function as an evolving, integrated organism rather than a static structure.

Industrial construction could also be automated through the process of continuous frame structures consisting of metal, glass, plastic and reinforced, pre-stressed, lightweight concrete. These would probably be selected as universal units for constructing factories, educational facilities, harbor systems, etc. Mega-machines could construct entire buildings by computerized, pre-programmed instructions, reducing construction time considerably. This programming could be readily altered to fit changing conditions.

The architecture and individual dwellings of the future would evolve in a completely different manner from today's structures. With the intelligent application of humane technologies, we could provide a wide array of uniquely individual homes. Structural elements would be flexible and coherently arranged to best serve each individual. These prefabricated, modular homes, embodying a high degree of flexibility inconceivable in times past, could be built in any place one might imagine, whether the preference be amidst forests, atop mountains, or on remote islands.

All these dwellings can be designed as self-contained residences, with thermal generators, heat concentrators and photovoltaic arrays built into the skin of the building, and with thermo panes to tint out the bright sunlight by variable patterns of shading. All these features can be selected by the occupant, and can supply more than enough of the energy required to operate the entire household.

Buildings may also contain a precise combination of dissimilar metals utilizing the thermocouple effect for heating and cooling, and other materials embedded in solid state plastic or ceramic, structural materials. The warmer it gets on the outside, the cooler it becomes on the inside. The application of this principle can serve to heat or cool the buildings. The interiors of the homes would be designed to suit the preferences of the individuals.

TRANSPORTATION

In the new, cybernated cities of the future personal travel to distant locations may be less desirable because of the many interesting events, options, and activities immediately available near home.

When travel outside the city is desired or necessary, computer-guided vehicles for land, sea, air, space and beyond could transport passengers and freight. With the adoption of integrated transport systems passengers and freight would be moved with a minimal expenditure of energy.

For rapid movement of passengers on land across viaducts, bridges and tunnels, high-speed mag-lev trains could span great distances and replace most aircraft transportation. Some of the passenger compartments in the transport units can be lifted from the moving train in transit, eliminating waiting time at stations. Rail, sea and undersea craft could handle most freight. Many of the transport units could have detachable components and contain internationally standardized containers, each of which could be easily transferred.

In the cities, types of escalators and elevators, conveyors and transveyors, move in all directions while interconnected with all other transport systems. A circular route around the outer perimeter would connect conveyor systems to radial and vertical extensions, making it possible to travel to any part of the city quickly. Future cities can be designed to accommodate all transport systems for the convenience of passengers and freight alike.

Most of the smaller transportation units for people could be operated by voice control. When voice control is not practicable or possible, alternative methods such as keypads could be used. All transportation systems would be modular, continuously updated, and provided with the latest developments in current technology.

The transportation system within the cybernated cities of tomorrow extends to the home as well. Although all manner of goods and services would be readily available in the city center, most people, if they chose, could access materials and information from their places of residence.

CONSPICUOUS WASTE

If our civilization is to endure, it must eventually outgrow the need for conspicuous waste of time, effort, and natural resources. This may be seen in any number of areas. At one time, architectural adornments were an integral part of construction. The lofty columns and colonnaded porticos of ancient Greece and Rome were necessary components of the structures. But with the advent of newer, lightweight materials and engineering improvements, we now span greater distances without these columns or other intervening support structures. Yet the designers of many of our government buildings, including the Capitol in Washington, D.C., engage in the conscious withdrawal of efficiency to maintain designs felt to be impressive, but which actually reflect mere convention and artificiality.

Designing a building with many projections of artificiality does not indicate originality, creativity, or individuality. Individuality is expressed in our unique way of thinking about ourselves and the world around us, not in our external appearance. If we continue to design our buildings with conspicuous waste and decoration, we lessen the standard of living for others.

This is not to detract from the beautiful structures created in the past with the available and limited technology of the times. However, the continuing application of these ancient methods of construction retards the innovative and creative thinking which is necessary to an emergent culture. The intelligent use of resources incorporated into structures will considerably simplify our lifestyle and reduce waste and maintenance.

These new cities will provide for the needs of the inhabitants through the efficient allocation of resources and materials, in an energy-conscious and pollution-free environment.

In human systems, evolution distributed eyes and internal organs in a fairly uniform manor. The same is true for all plant and animal species. Uniformity is not necessarily a bad thing if it functions towards a satisfactory end. However, the dangers of uniformity are evident in our inability to shrug off useless values or methods which have outgrown their usefulness. Perhaps the only kinds of uniformity acceptable in the future will be the protection of the environment and concern for our fellow human beings.

We have to ask ourselves what kind of world we want to live in: the choice and responsibility are ours.

HOLISTIC CONSIDERATIONS IN A RESOURCE-BASED ECONOMY

In a resource-based economy, holistic considerations become an integral part of overall planning. A careful investigation of the positive and negative effects of each project would be carefully analyzed before any project is undertaken.

The cities we propose offer the possibility of tremendous intellectual growth, with its emphasis on environmental and human concern. These cities would be free of noise, pollution, most crime, and many other deleterious conditions associated with the cities of today.

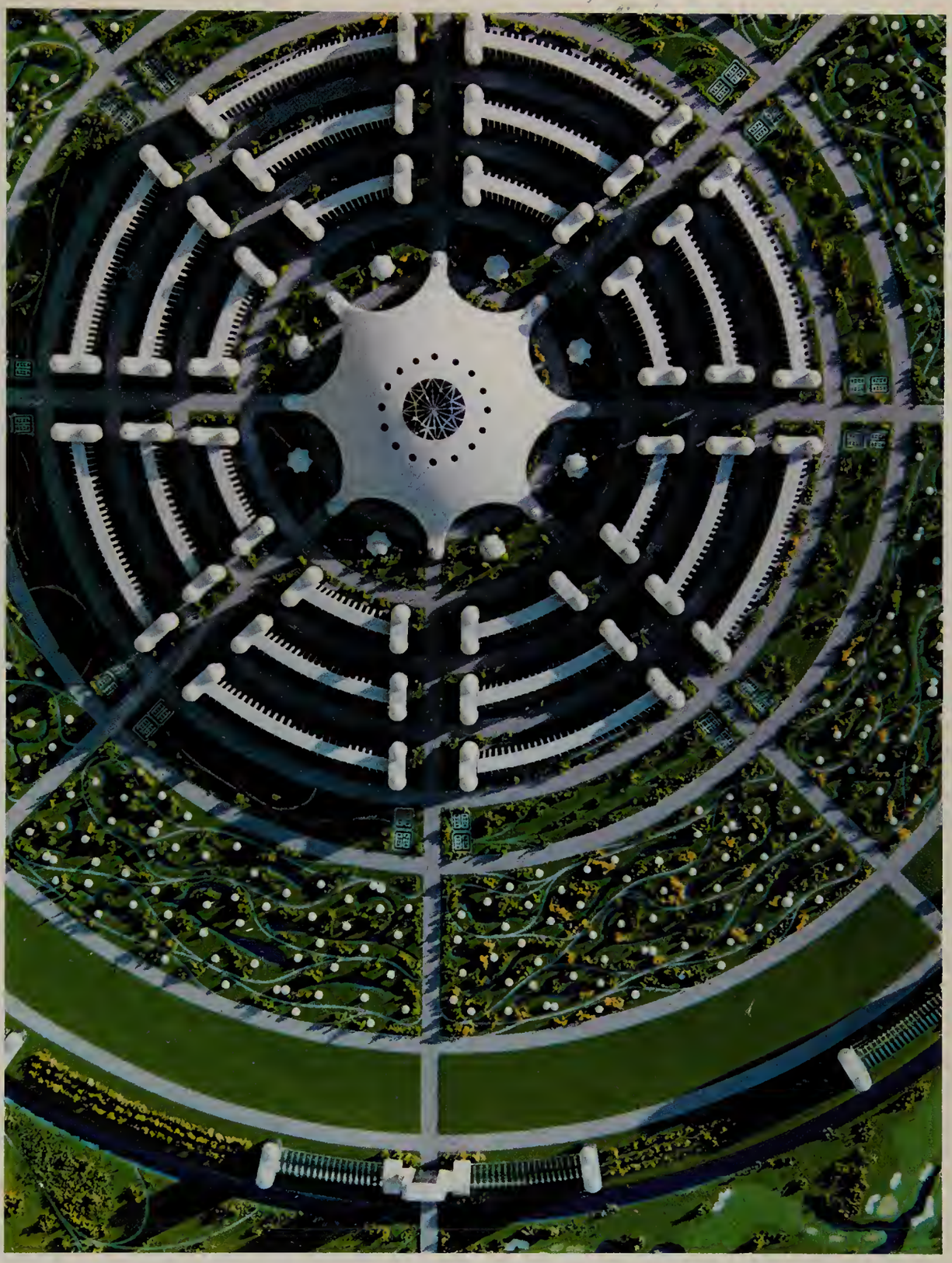
Of course, people will be free to live wherever they choose. But these cities are planned with plenty of open country, parks, and wooded areas. In the areas for individual housing there will be enough vegetation and trees between houses to impart a sense of privacy.

At first glance, our proposal for a city of the future may appear impracticable; yet it represents an achievable, sustainable, and sophisticated environment, one that is designed to help bring out the best in the human potential. These cities will not only provide resources

and information but will be university cities of continuous growth, designed to encourage individuality, creativity, and cooperation — with concern for the total person.

The transition to this social arrangement will not be an easy one. Never in human history has there been a smooth transition from one social system to another. Any major change engenders a certain degree of resistance. The most effective way to implement such a change is through worldwide use of media, seminars and workshops during the initial design stage

In the final analysis, talk proves little. Just as all new ideas go through a process of maturation and development, we expect our experimental City of the Future to gradually gain acceptance by fulfilling its promise as a highly successful, peaceful, and desirable place to live. As newer communities develop and become widely accepted, they will provide the basis of a new society through the process of evolution rather than revolution.



CIRCULAR CITY

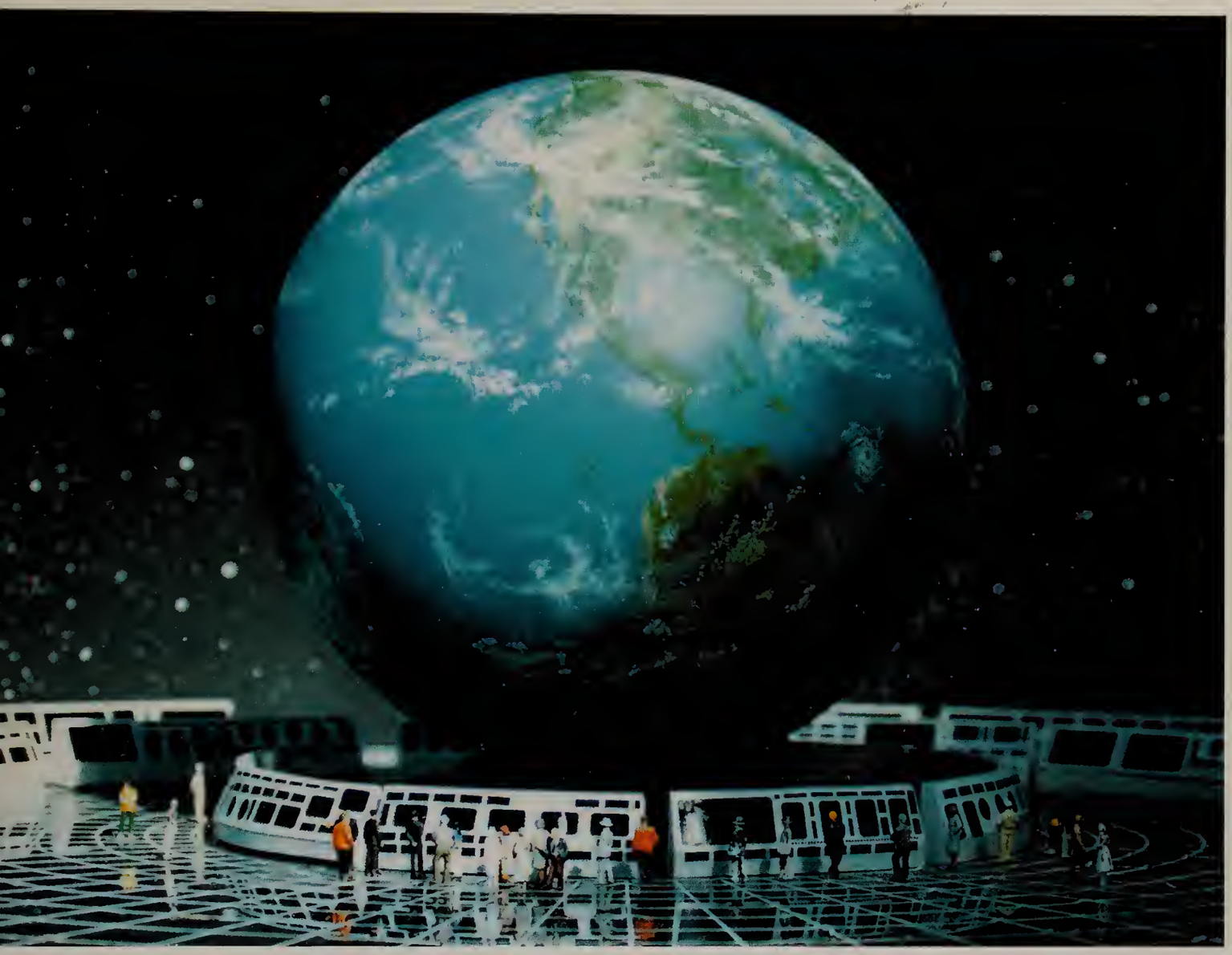
The outer perimeter will be part of the recreational area with golf courses, hiking and biking trails, and opportunities for water sports. A waterway surrounds the agricultural belt with its enclosed, transparent buildings. The application of newer technologies will eliminate once and for all the use of dangerous chemicals and pesticides. Continuing into the city center, the eight green sectors provide clean renewable sources of energy with wind, thermal, and solar energy devices. The residential belt features beautiful landscaping, lakes, and winding streams. The homes and apartments will be gracefully contoured to blend in with the landscape. A wide range of innovative architecture will provide many choices for the occupants.



CIRCULAR CITY

Adjacent to the residential district, a wide selection of healthy, organically-grown food will be available on a 24-hour basis. Next are the apartments and design centers. Surrounding the central dome, eight domes house the science, art, music, research, exhibition, entertainment, and conference centers, which are all fully equipped and available to everyone.

The central dome or "theme center" houses the cybernated system, educational facilities, a health center, as well as facilities for shopping, communications, networking, and childcare. In addition, it serves as the core for most transportation services, which will take the form of horizontal, vertical, radial, and circular conveyors that safely move passengers anywhere within the city. This system facilitates efficient transportation for city residents, eliminating the need for automobiles. City-to-city transportation would be provided by monorail and electrically operated vehicles.



CYBERNATED COMPLEX

This cybernated complex utilizes advanced imaging technology to project a 3-D, "virtual" image of the earth in real time. It utilizes satellite communication systems to provide information on worldwide weather conditions, ocean currents, resource inventories, population, agricultural conditions, and fish and animal migration patterns. The interconnected, cybernated complexes represent the brain and nervous system of the entire world's civilization. All of this information will be available on demand to everyone via the Internet. This single site manages our common heritage — the resources, carrying capacity and health of the earth.



UNIVERSITY CITY

This University of Architecture and Environmental Studies, or "World University," is a testing ground for each phase of architectural development. This would be a "living," continually evolving research institute open to all. Student performance would be based on "competence accreditation" and research findings would be applied directly to the social structure to benefit all of humanity.

People will live in these experimental cities and provide feedback on the livability and serviceability of the various structures. This information would be used to formulate modifications to structures so that maximum efficiency, comfort, and safety are assured. This facility is also used to develop modular construction systems and components that serve a wide range of needs and preferences. In most instances, the external appearance of the buildings will reflect the function of the building – they are designed "from the inside out."



MILE-HIGH SKYSCRAPERS

These skyscrapers are constructed of carbon fiber-reinforced and pre-stressed concrete. They will be stabilized against earthquakes and high winds by three massive, elongated, tapered columns, which are 100 feet wide and almost a mile high. This tripod-like structure is reinforced to diminish compression, tension, and torsion stresses. These super-sized skyscrapers assure that more land will be available for parks and wilderness preserves, while concurrently helping to eliminate urban sprawl. Each one of these towers encloses a complete environment, containing a shopping center as well as childcare, educational, health, and recreational facilities. This will help alleviate the need to travel to outside facilities. Although the author is not in favor of mile-high skyscrapers, if we do not maintain a balance between the population and the earth's carrying capacity we may have to move our cities not only skyward and seaward, but subterranean as well.

CENTER FOR DIALOGUE

The directive of The Center for Dialogue would be to submit the urgent issues of the times to critical examination, and to raise relevant questions for informed public dialogue.

BUILDING OF DOMED STRUCTURES

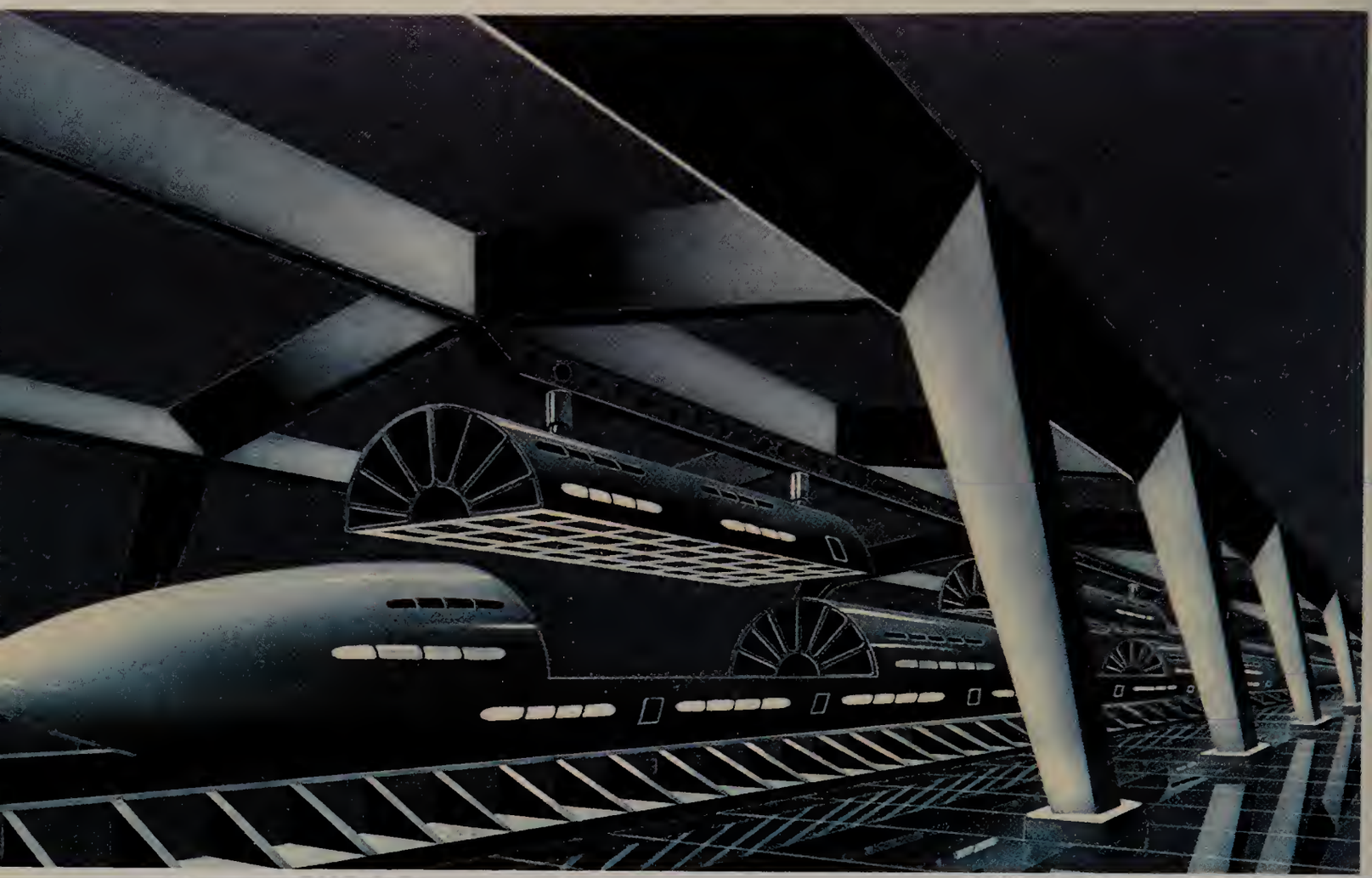




BRIDGES

These elegant bridges are designed to carry compression, tension and torsion loads in the simplified expression of its structural members. Mag-lev trains are suspended beneath the enclosed traffic lanes.



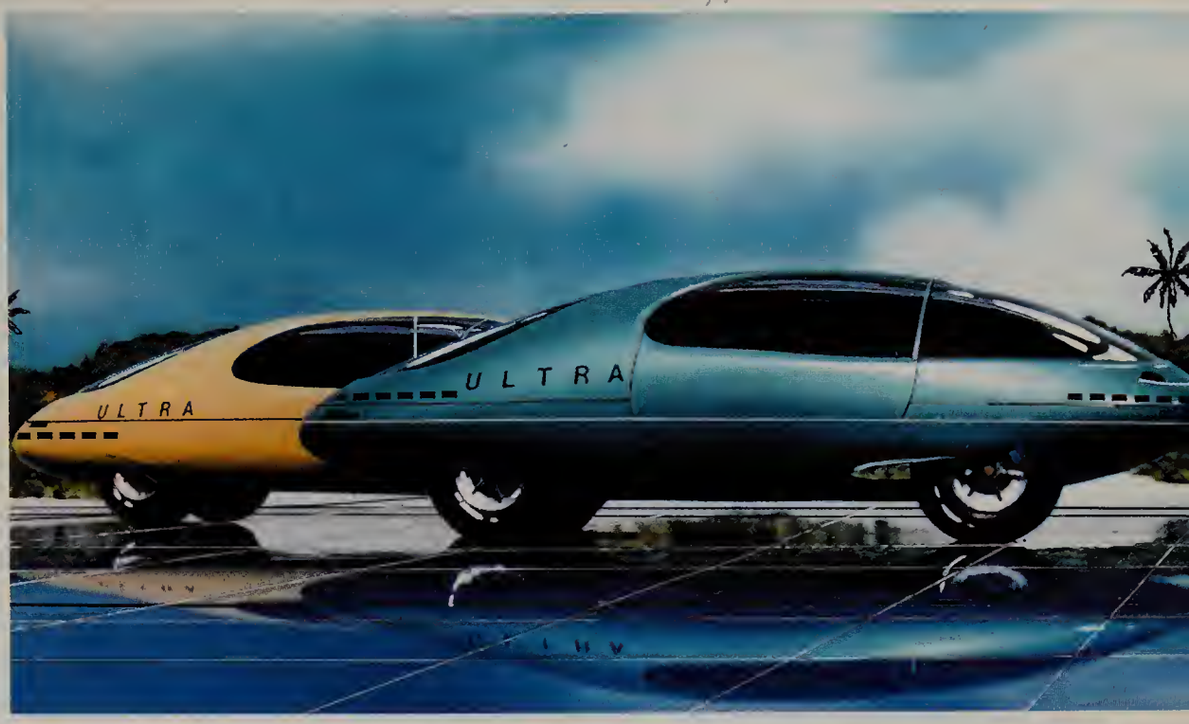


MAG-LEV TRAINS — MASS TRANSPORTATION SYSTEMS

While these high-speed, magnetic levitation trains are in motion, a segment of the passenger compartment can be either lifted or slid to the side. These detachable sections can then take passengers to their local destinations while other compartments are lowered in their place. This method allows the main body of the train to remain in motion saving time and enhancing efficiency. In addition, the removable compartments could be specially equipped to serve a wide range of transportation services.

AUTOMOBILES

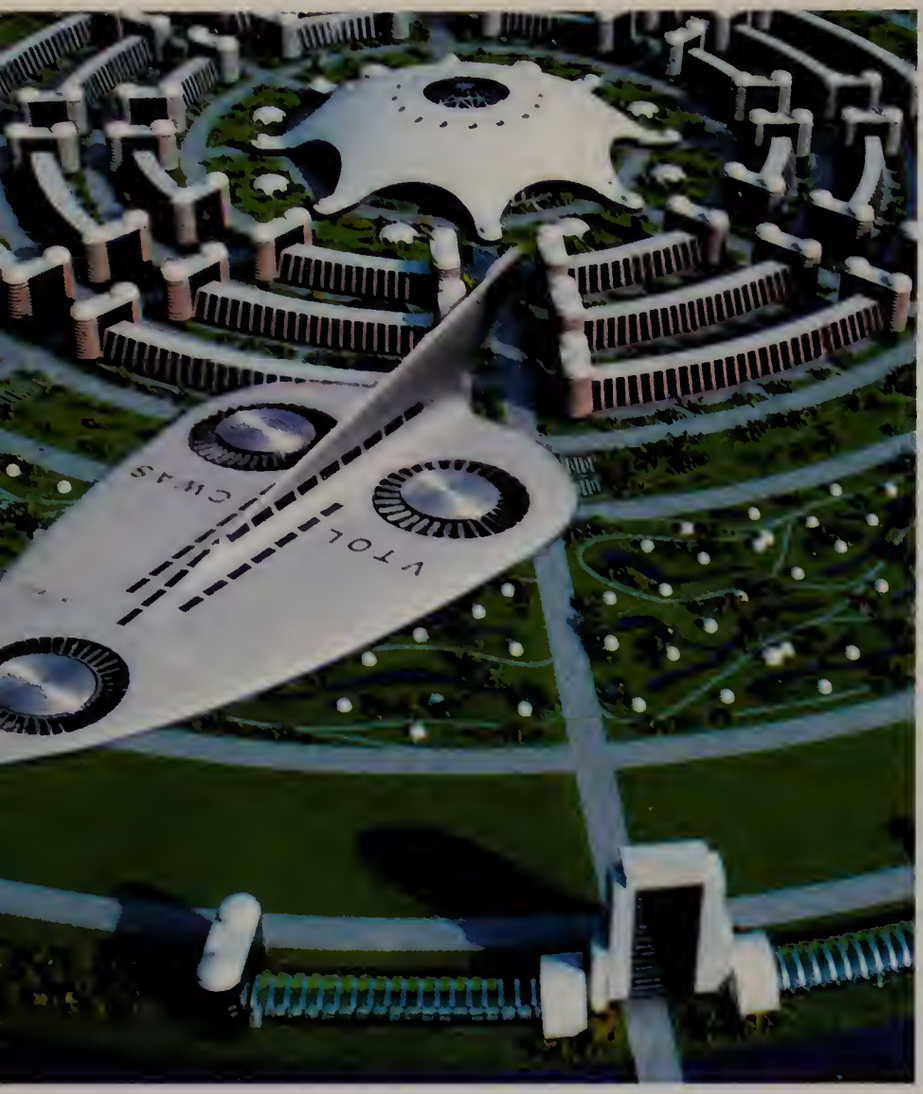
Streamlined cars provide high-speed, energy efficient, and safe long-range transportation. Some vehicles will have wheels, while others will eventually be equipped with magnetic levitation or air-floatation capabilities. Most vehicles will be equipped with voice-recognition technology that allows the passengers to request their destination by voice command. Self-monitoring systems tell the vehicles when service is required, and they transport themselves to service and maintenance facilities. Use of clean, non-polluting electrical energy allows for silent vehicle operation, while proximity-sensor devices linked to automated velocity and braking systems enhance safety by enabling the vehicles to avoid collisions. As a secondary safety measure, the entire interior will be equipped with ergonomically-designed air bag systems. Within the cities, horizontal, vertical, radial, and circular conveyors will serve most transportation needs.





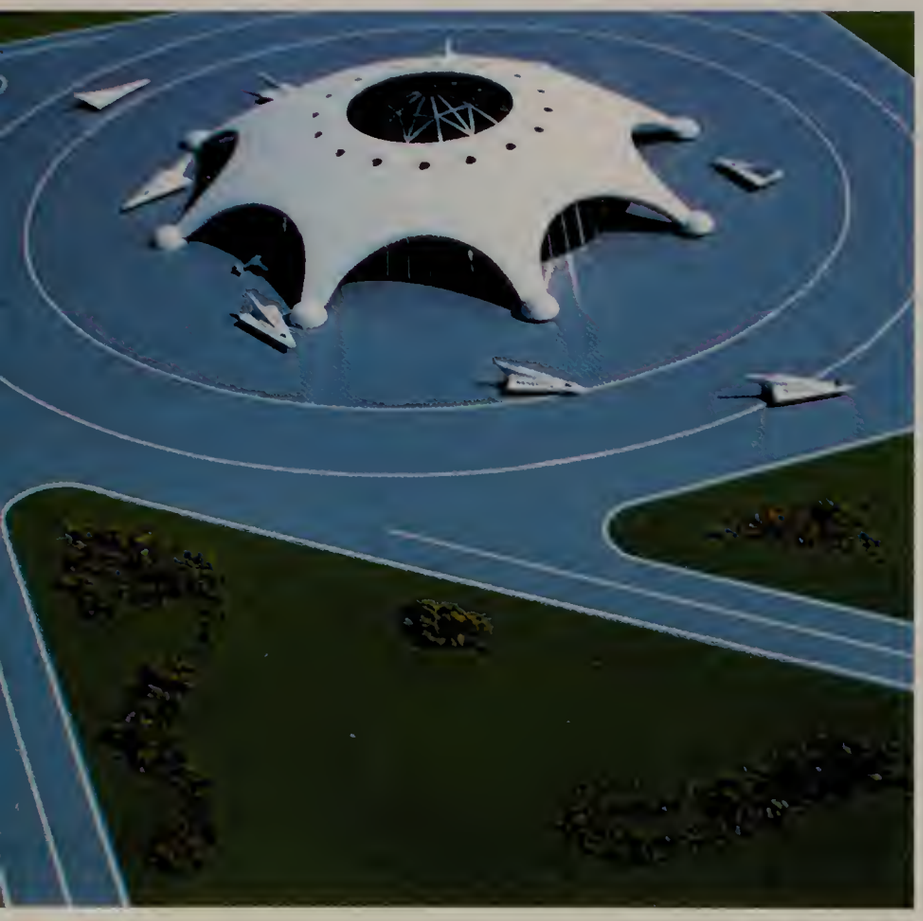
AIR TRANSPORTATION

These Vertical Takeoff and Landing (VTOL) aircraft lift passengers and freight by the use of ring-vortex air columns. The helicopter in the foreground has a stationary center around which the rotors are propelled by engines at their tips. VTOL aircraft will be propelled by a variety of techniques, from ducted fans to vectored jets. They will be designed to combine the most desirable attributes of fixed-wing aircraft, helicopters, and flying platforms. Transcontinental travel will be achieved through advanced aircraft and high-speed mag-lev trains, all integrated into a worldwide transportation system.



FUTURE AIRCRAFT

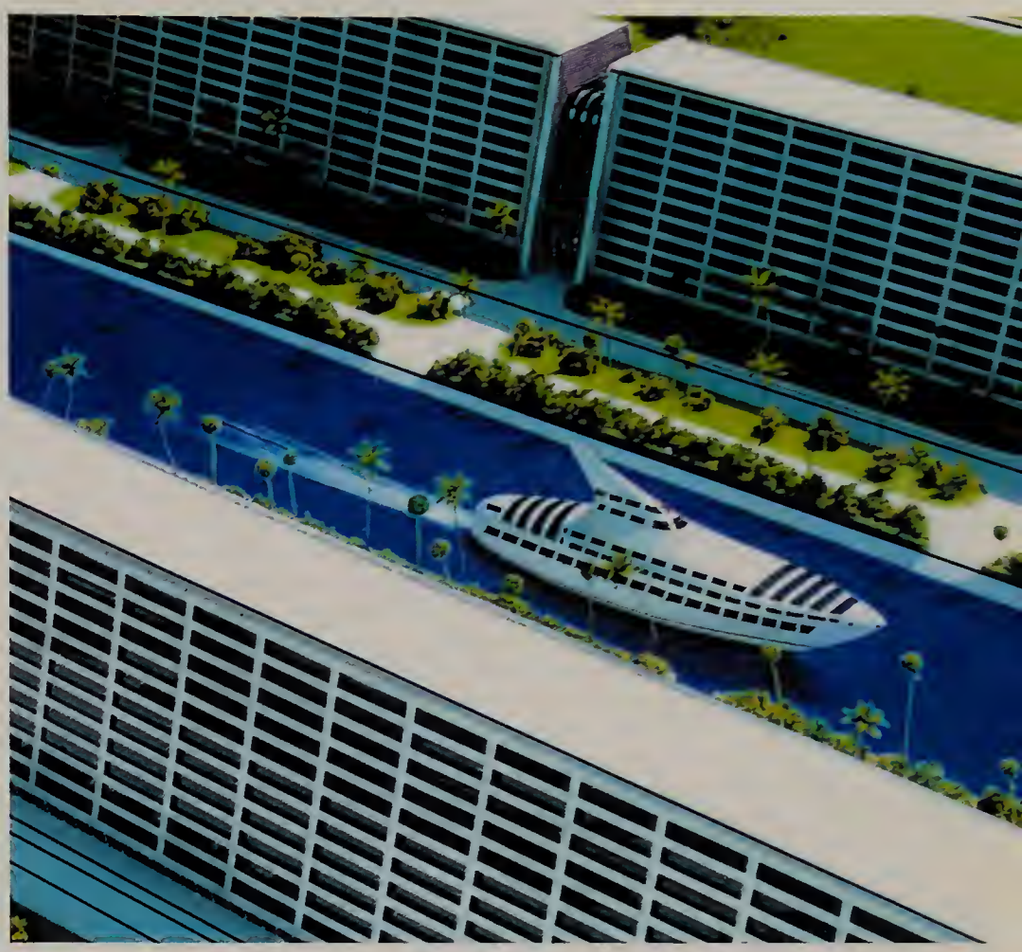
Since military aircraft will be unnecessary in the future, emphasis can be shifted to advancing medical, emergency, service, and transportation vehicles. Here is an example of VTOL (Vertical Take-off and Landing) aircraft with three synchronous turbines, which allow exceptional maneuverability. These delta-configuration aircraft can be controlled by electrodynamic means, eliminating the need for ailerons, elevators, rudders, spoilers, flaps or any other mechanical controls. In addition to providing better maneuverability and aerodynamic qualities, this innovative technology also includes a de-icing system. In the event of an emergency landing, fuel will be ejected to prevent fires.



AIRPORTS

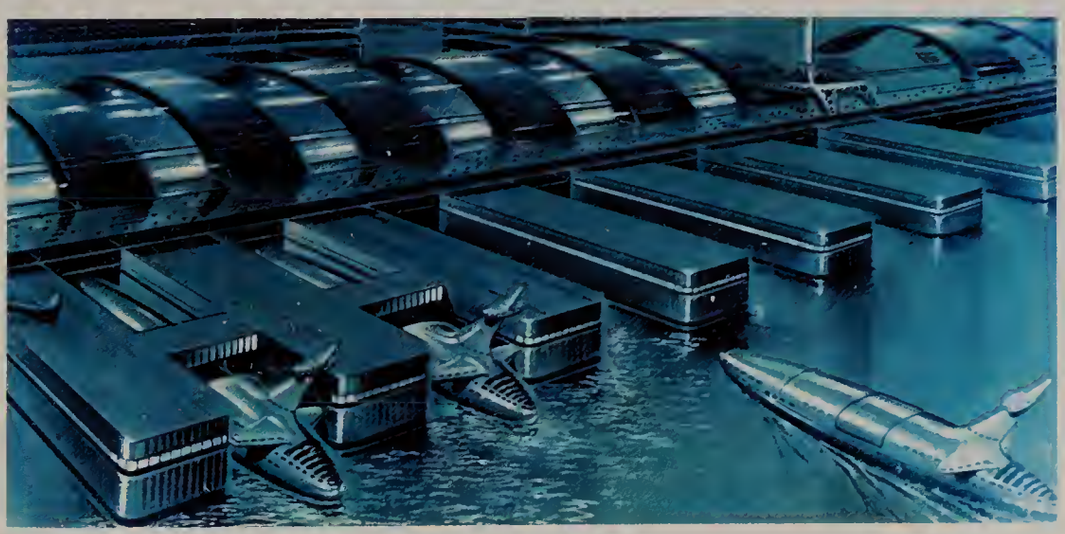
The central dome of this airport of the future contains air terminals, maintenance facilities, service centers, and hotels. The runways are arranged in a radial configuration, which allows airplanes to easily take off into the prevailing winds and to avoid dangerous crosswind landings. Emergency stations at the edge of the runways are fully equipped with built-in fire fighting equipment and emergency arresting gear. All of the runways will be equipped with built-in sprinkler systems. Passengers will be transported to and from the airport by underground conveyors. Many of the terminals themselves will eventually be constructed underground for increased safety and better use of the land.

AUTOMATED CANALS & WATERWAYS



Included in a national transportation system would be a network of waterways, canals, and irrigation systems. We can no longer treat natural and man-made elements of the environment as stand-alone systems. These "mega hydrological projects" will be an integrated part of intercontinental planning. These bodies of water could minimize the threat of floods and droughts while allowing the migration of fish, removal of accumulated silt, and creation of sites to manage and "clean" agricultural and urban run-off. These waterways would also be part of an international flood control system that diverts floodwaters to water storage basins, allowing the water to be utilized during periods of drought. This would not only help maintain the water table, but would also provide natural firebreaks as well as an emergency water source for fires. In addition, these canals would supply water for farming and irrigation, supply the nation's land-based fish farms, protect the wetlands and wildlife, and supply water to recreation areas.

INTERNATIONAL SHIPPING SYSTEMS



These ships will be in effect floating, automated plants, capable of processing raw materials into a finished product while en route to their destination. Some serve as industrial fish processing plants and canneries, while others would be equipped with multi-cellular compartments capable of transporting a wide variety of products.

Hydrodynamic seafaring vessels permit high-speed, efficient travel. They will be energy efficient and provide maximum comfort and safety for all passengers. They will be manufactured from durable composite materials, their outermost skin consisting of a thin layer of titanium, which requires minimal maintenance. Portions of the upper deck will slide open when the weather permits.

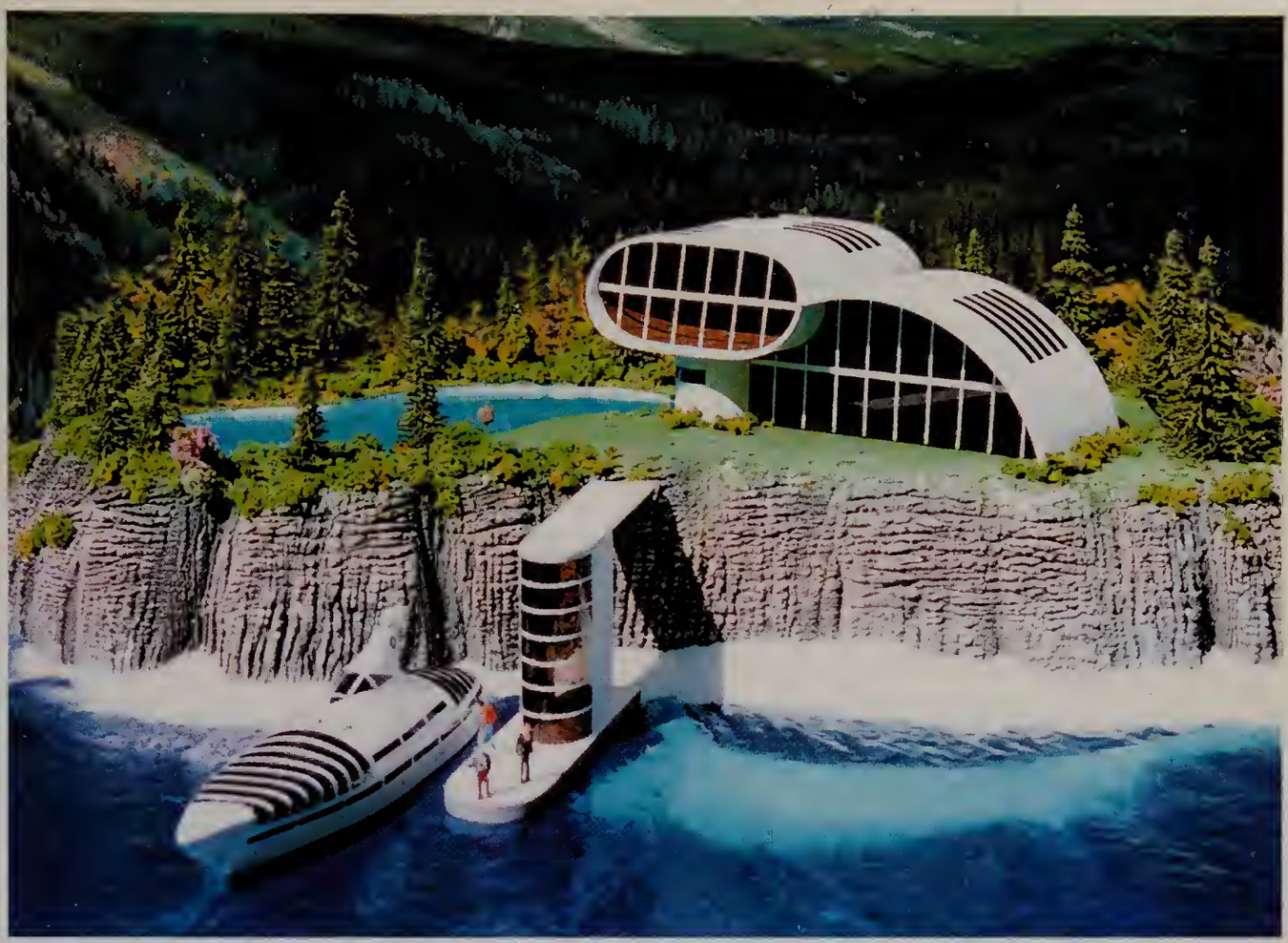






HOMES

The architecture and individual dwellings of future cities will evolve on an entirely different basis from today's houses. With the intelligent application of humane technologies, we will be able to provide and allow for a wide array of unique individual homes. Their structural elements will be flexible and coherently arranged to best serve individual preference. These pre-fabricated, modular homes, embodying a high degree of flexibility inconceivable in times past, could be built in any place one might imagine, amid forests, atop mountains, or on remote islands.





HOMES

Homes could be prefabricated of a new type of pre-stressed, reinforced concrete with a flexible ceramic external coating; they would be relatively maintenance-free, fireproof, and impervious to the weather. Their thin shell construction can be mass-produced in a matter of hours. With this type of construction, there would be minimal damage from earthquakes and hurricanes.







HOMES

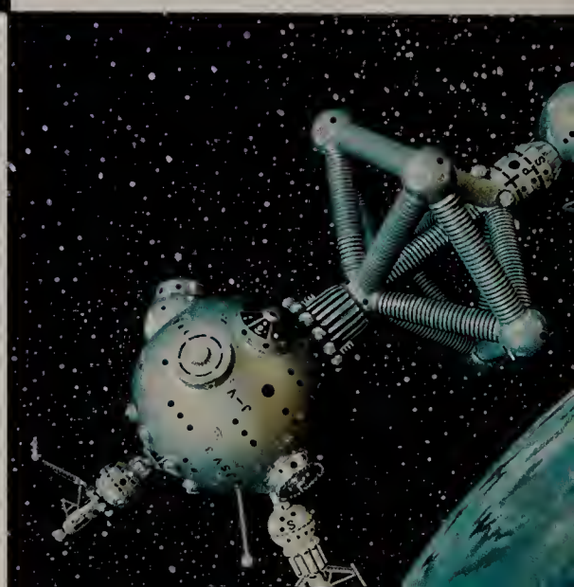
All these dwellings can be designed as self-contained, energy efficient residences with their own thermal generators and heat concentrators. Photovoltaic arrays would be built into the skin of the building and into the widows themselves. "Thermopanes" tint out the bright sunlight by variable patterns of shading. All these features could be selected by the occupant to supply more than enough of the energy required to operate the entire household.





SPACE STATIONS

Space stations provide the advantages of a gravity-free research environment. They can be entirely automated and self-contained to permit maintenance and self-repair without human intervention. These space stations would be able to monitor the earth's resources, as well as facilitate further research in the fields of meteorology and astronomy, work that is often difficult on Earth due to atmospheric interference. Many other experiments can be accomplished in a gravity-free environment, particularly in the areas of medicine, chemistry, and metallurgy. In addition, these space stations would serve as nodes in a worldwide telecommunications system, providing up-to-date information on the Earth's ecosystems, the position of ships and airliners, and other information pertinent to the inhabitants of the cybernated world.



CHAPTER 16

LIFESTYLE IN THE FUTURE

FAMILY MATTERS

THE MEDIA AND POLITICIANS TALK A LOT about the dissolution of conventional family structure, and the societal values associated with it. They see the family as the primary, most basic venue for acquiring such life skills as sociability, responsibility, stability, and concern for others. The increasing unrest and lack of direction exhibited by many young people seem to validate these concerns for them.

At present, it is necessary for both husbands and wives to work. Monetary economics have to a large extent undermined family cohesion and childcare. Parents lack adequate time to spend with their children, and they are constantly stressed by ever-rising medical bills, insurance payments, educational expenses, and cost of living expenses. Ironically, for many, one of their most pressing expenses is childcare, to pay for which they must go to work.

It is in this area that one of the most profound benefits of this new civilization could be realized. Shorter workdays will provide greater opportunities for family members to peruse their areas of personal interest. Free access to goods and services will make the home a much more pleasant place, with the removal of the economic stress that causes so much family turmoil.

The question arises: Would people be happier in this kind of society? Perhaps it isn't so much happiness that we seek: happiness is relative to each individual's distinct nature, and is thus individually defined and achieved. We seek to create a society where people are free to choose their own areas of interest, develop formerly hidden potential, and pursue their dreams without government intervention or financial constraint.

CHANGING VALUES IN A CYBERNATED SOCIETY

As of this writing, the vision of the good life in industrial nations, particularly the United States, entails productive and rewarding employment, a healthy existence, perhaps a home in the suburbs and a second in the country, money in the bank, good education, pleasant family relations, healthy and intelligent children, and the guarantee of a secure future. This projected fantasy is an elusive goal for almost all people.

Tomorrow's vision of the good life may differ considerably. With the elimination of scarcity, and when most material needs are met by a world resource-based economy, we may reasonably expect to see considerable improvement in living conditions and opportunities for a meaningful and productive life for all people.

Consider for the moment what the world could be like if most physical needs are met. What will happen to individuality and human values in a world of unlimited abundance?

Rather than evolving into an age of dynamic leisure, truly intelligent and committed people find very little free time, even if they don't have to "work" for a living. The energies, once devoted to dealing with the myriad problems of material scarcity, could now be directed toward self-development and fulfillment. People would then have the means and the time for intellectual and spiritual growth, and the time to realize what it really means to be human in a caring society. When social affairs are consistent with the carrying capacity of the earth's resources, human beings will evolve a sense of relevance and understanding far beyond that which is possible today.

All people are culture-bound. We are victims of indoctrination and of our social customs. Most of us would be bewildered and even uncomfortable with the flexibility of a new orientation. Today most of us live in economic and mental straitjackets that limit our ability to work through our problems. For the first time in human history the cybernated world offers us the opportunity to choose whatever constructive lifestyles we wish. All individual lifestyles would be determined by one's varied and changing preferences, and not by what someone else thinks is good for one.

An example of the wide range of choices that may be available in a resource-based economy would be in the way one selects a house. For instance, a man and woman may visit an architectural design center where they sit in front of a clear hemisphere approximately six feet in diameter. The woman describes the type of house she would prefer to live in and her areas of interest. The house appears as a 3-dimensional image in the center of the hemisphere. It rotates slowly to present an overview of the interior and exterior. Then the man describes his major areas of interests and preferences and suggests a larger balcony. The 3-dimensional image is adjusted. When they have finished requesting changes, the computer presents various alternatives to consider. Also, if they wish, they can enter a sensorium to experience a walk-through of their preferred design and continue to make changes.

When they arrive at the desired design, then the construction procedures are set in motion. The computer selects materials for efficiency and durability. None of the architecture is permanent and can be modified and updated at the request of the occupants.

This is real, individual choice. In a monetary system, most of us live near our work with a house, car and lifestyle we can afford (or, all too often, cannot afford), rather than the one we prefer. The truth is, we are essentially only as free as our purchasing power permits. Even many wealthy people today select a residence solely to impress others with their status. Not having a true sense of self worth, many live to impress others.

A resource-based economy changes the function of our dwellings from that of status symbol or basic shelter to a reflection of our individuality and interests.

A resource-based economy would provide art centers, music centers, theater projects, and an opportunity for all people to return to an educational environment, travel, and pursue their interests. Although people would be economically secure, they would still need real challenges to maintain incentives and enhance creativity. The future will offer these challenges in abundance.